

AI Assistance in Medical Decision-Making: The Role of Recommendations and Explanations in Simulated Clinical Cases

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Fig. 1. The clinical process and multi-stage decision points.

Healthcare is a sector that can greatly benefit from Artificial Intelligence (AI) due to the critical nature of medical decisions and the demand for accuracy and efficiency. Advances in AI algorithms have shown promise in diagnosing conditions at early stages. To improve the transparency of such systems, Explainable AI (XAI) techniques have been explored. However, if not properly integrated, AI and XAI can introduce new challenges, such as reduced cognitive engagement or over-reliance on AI system outputs. While prior research has mainly focused on isolated tasks, such as AI-assisted diagnosis, this study takes a *holistic* approach by examining the impact of AI recommendations and explanations across multiple sequential decision points within clinical case scenarios.

Thirty-four medical students completed two simulated clinical cases with varying levels of AI assistance. The scenarios reflected realistic workflows, requiring participants to prescribe diagnostic exams and make final diagnoses. Results indicate that AI recommendations and explanations can enhance medical performance, but do not consistently improve efficiency. In simpler cases, AI assistance supported appropriate exam prescription and reduced decision-making time. In more complex cases, explanations significantly improved diagnostic accuracy but also led to higher resource use, with participants ordering more unnecessary exams.

CCS Concepts: • **Human-centered computing** → **User studies**; • **Computing methodologies** → **Artificial intelligence**; • **Applied computing** → **Health care information systems**.

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1 Introduction

In recent years, AI has rapidly emerged as a transformative technology. It is reshaping various industries and forms of work by leveraging its ability to analyse vast datasets and generate predictive models to support better decisions [27]. Sectors such as finance, criminal justice, and healthcare have seen significant advancements driven by AI and Machine Learning (ML) algorithms by optimising processes, enhancing outcomes, and driving innovation [25].

Despite the rapid advancements in AI, its integration into real world applications, particularly in high-stakes environments like healthcare, has been slower [11]. In healthcare, where accurate and reliable decision-making is essential, concerns arise regarding the effectiveness of AI implementation as each decision must be accurate and precise and most often than not AI systems work as a black box in which is difficult to assess their reasoning [50]. Explainable AI (XAI) techniques have been developed into several different sectors and type of tasks to make AI systems more transparent and trustworthy, enabling users to better understand and interpret AI recommendations [2, 50]. For example, in skin cancer diagnosis, heatmaps are used to visually explain AI predictions by highlighting areas of the image relevant for classifying a lesion as malignant or benign [23]. However, the standardisation of these technologies remains unsettled, particularly in selecting the most appropriate explanation methods for different tasks and contexts [10, 54]. XAI methods in healthcare must be tailored to specific tasks. Heatmaps are useful for image-based diagnoses, while feature importance scores are more appropriate for tabular data, where understanding the influence of variables is crucial. This variety in data types and tasks complicates the development of a standardized XAI approach, as no single method suits all scenarios, [10].

The full integration of AI into healthcare remains uncertain. Although there is ongoing research into incorporating AI into medical diagnosis and clinical practice [36, 48, 61], our understanding of how medical professionals interact with these systems in real-world decision-making scenarios is still limited. Prior studies have primarily examined AI support in *isolated medical tasks*, such as image interpretation, diagnostic classification, or treatment prediction [13–15, 29, 52, 60], with little attention paid to its influence across the broader clinical decision-making process. Yet medical reasoning typically unfolds as a sequence of interdependent decisions: (1) ordering diagnostic exams¹ based on an initial hypothesis, (2) determining the most likely diagnosis from exam results, and (3) selecting an appropriate treatment plan. We refer to the examination of these interconnected stages as a *holistic* perspective on clinical decision-making, emphasizing the reasoning process across sequential tasks rather than isolated tasks. Within this lens, it is necessary to evaluate how AI recommendations and explanations affect performance and efficiency across multiple, sequential decision points within the same case scenario. Therefore, we investigate the following research questions:

- RQ1.** How do AI recommendations and explanations influence diagnostic accuracy and exam prescription appropriateness throughout the holistic clinical assessment process?
- RQ2.** How do AI recommendations and explanations impact cumulative decision-making efficiency (time and resource use) across sequential clinical stages?
- RQ3.** How do these effects vary with case complexity within a holistic decision-making framework?

We conducted a user study with medical students who were tasked with solving two clinical case studies using a simulator platform. In the study, we manipulated the level of AI assistance provided to solve the clinical case studies: one group of medical students received no AI assistance, another group received AI recommendations, and a third group received both AI recommendations and detailed explanations. Our findings suggest that while

¹Although *tests* usually denote diagnostic or laboratory procedures and *exams* refer to clinical or physical evaluations, in this study the two terms are used interchangeably to denote diagnostic procedures requested by the clinician.

AI recommendations and explanations improve medical performance in terms of the accuracy of prescribed exams and diagnosis, they do not guarantee efficiency. Moreover, presenting explanations increased reliance on the AI agent.

By simulating real-world clinical decision-making, this study examines how AI recommendations and explanations influence physicians across multiple key sequential decision points. This approach provides a structured way to assess medical behaviour in clinical assessments under varying conditions. The main contributions of this paper are:

- C1.** We investigate AI-assisted medical decision-making across multiple key decision points within realistic clinical workflows, offering a novel structured analysis of AI's influence on the decision-making clinical process.
- C2.** We examine the impact of AI recommendations and explanations in clinical cases of varying complexity, providing insight into how the difficulty of the case shapes AI-assisted medical decision-making.
- C3.** We evaluate medical efficiency, as a critical yet often overlooked metric in AI-assisted medical decision-making, analysing how AI influences both the speed and resource use of clinical assessments.

We investigate AI-assisted medical decision-making across multiple key decision points, providing a *holistic* analysis of AI's influence throughout the diagnostic process.

By adopting a multi-stage, process-oriented perspective, this work offers a more realistic account of how clinicians interact with AI systems in practice, uncovering critical trade-offs between diagnostic accuracy and resource use that prior single-task investigations could not capture.

2 Related Work

2.1 Human-AI Interaction Challenges

As AI systems become more prevalent across various sectors, understanding how humans interact with these technologies is increasingly important. While the development of AI algorithms has progressed rapidly, research into how people adapt to and trust these systems hasn't kept the same pace [56]. The complexity of this interaction lies in the reluctance of individuals to adopt AI because of the challenges it brings. Firstly, because AI systems are non-deterministic, the outcome of their actions can be unpredictable, which may lead to apprehension [62]. Secondly, people are expected to collaborate with AI as teammates, complementing each other's knowledge and capabilities, but sometimes the most accurate AI algorithms do not always result in the best team performance [6, 7]. Thirdly, there is a concern about the opacity of AI systems, which are often viewed as "black boxes". Users may not understand how and why AI systems make decisions or know when to trust them.

To address these challenges related to fear and understandability, XAI has emerged as a critical approach. XAI techniques are designed to make AI predictions more transparent and comprehensible by providing explanations that clarify how decisions are made [59]. In fact, explanations have shown to actually foster trust and confidence in AI systems recommendations, [12, 44]. However, these explanations can sometimes fall short in practice, as they may introduce cognitive biases [9, 18], leading to overreliance. This can occur when individuals develop overconfidence in AI suggestions merely because an explanation is provided or when explanations reduce critical thinking [19, 20, 24]. For example, feature importance explanations, which highlight which features influence a model's predictions, can be difficult for those without a background in mathematics to understand [24]. These explanations often rely on abstract representations or statistical metrics that may not be intuitive to users who are not familiar with the underlying mathematical concepts. Current XAI approaches, particularly in the healthcare sector, often fail to address cognitive biases effectively, revealing a gap between the intended transparency of XAI and its practical applicability [24]. This underscores the need for XAI techniques that are not only theoretically sound but also practically useful and accessible to a broader audience. Recent research explores aligning XAI

outputs with the needs of domain experts by translating technical explanations into formats that clinicians can interpret [36], but this work is still in its early stages and has not been extensively studied.

Recently, Large Language Models (LLMs) have emerged as promising tools for enhancing explainable AI (XAI). As advanced AI systems capable of natural language processing, LLMs can generate human-like text by understanding language and context, making their explanations potentially more intuitive than traditional XAI techniques. This has led researchers to explore their use as post-hoc explanation tools [33, 35], leveraging their ability to simulate human interaction to improve interpretability and user comprehension [63]. Building on this, studies have investigated different ways LLMs can enhance XAI. Some approaches transform outputs from traditional XAI techniques, such as SHAP [21] and LIME [46], into more natural and accessible narratives [67]. Additionally, interactive XAI methods have been developed, allowing users to engage with explanations through dialogue [26, 37, 41, 51, 53]. While explanations alone do not guarantee trust, LLMs provide new opportunities to improve user understanding and support more informed decision-making.

2.2 AI Integration in the Healthcare

Healthcare is one of the most prominent sectors for applying AI and XAI algorithms due to the critical need for precision and accuracy in medical decisions [4]. However, besides being a highly regulated environment, this sector's inherent conservatism aligned with the opaque nature of many AI algorithms contribute to scepticism among some healthcare professionals about adopting these technologies [8]. Especially because they worry that AI systems might cause them to doubt their decisions, which can lead to increased time spent on tasks [27]. Despite these challenges, AI has great potential to transform medical decision making, especially when combined with XAI techniques [32]. Studies show that AI can significantly improve diagnostic accuracy, addressing one of the most complex and demanding aspects of medical practice [13, 60]. Tschandl et al. [60], in a study on skin cancer diagnosis, and Cabitza et al. [13], in imaging exam interpretation, both demonstrated that AI, when working with clinicians, can improve diagnostic outcomes and outperform either clinicians or AI alone.

Beyond diagnosis, AI algorithms also enhance treatment planning, fostering trust through explanations, particularly for complex treatment decisions [44, 52]. However, several studies have indicated that while AI decision support can significantly influence prescription behaviours, explanations alone do not always improve clinical outcomes [39]. A study by Bussone et al. highlights risks associated with XAI in healthcare, such as overreliance when explanations are overly detailed or distrust when they are too sparse, both conducting to medical errors [12]. To mitigate these risks, they recommend presenting the system's confidence level alongside recommendations, which can help users appropriately calibrate their reliance on AI. Although this approach wasn't directly tested, other studies suggest that confidence scores can actually improve trust calibration [65]. To navigate the challenges of integrating explanations, Subramanian et al. [55] highlighted the importance of tailoring explanations not only to the specific task but also to the user's level of expertise, ensuring a balanced and effective use of AI in both diagnosis and treatment planning. Despite advancements, fully standardization of explanation techniques in healthcare AI is still challenging. Promising directions include novel XAI methods that create less doubts and space for incorrect interpretations [36], like generate human-like explanations and the use of LLMs that simulate peer-like conversations with clinicians [38]. These approaches aim to better engage users mimicking their interaction with peers and positively impact clinical decision-making, something that is currently being studied in the medical field.

The literature indicates that while recommendations and explanations have significant potential to improve healthcare, some challenges still remain in their effective integration. A major gap is the focus on isolated tasks, such as diagnosis [12, 13, 44, 60] or treatment [29, 52], without a complete view of the medical decision-making during clinical assessment and along multiple decision points. There is a need for further research on how physicians interact with AI throughout the entire clinical assessment process. Interacting with a task-specific AI

differs cognitively from working with a longitudinal AI assistant that contextualizes patient history across the entire clinical workflow. This shift may introduce specific efficiency and accuracy trade-offs, akin to those seen in decision-making under time pressure [57]. In this study we aim to understand AI's assistance impact in a broader way to explore physicians behaviour in a *holistic* perspective of the clinical assessment.

3 Study Design and Experimental Set up

3.1 The Task: Assessment of two Clinical Cases

In this study, we aim to explore the effects of AI recommendations and explanations in realistic clinical workflows where the AI system can be utilized at multiple decision points. A key requirement of the study is to simulate clinical assessments incorporating the three critical decision points in medical practice: **prescribing exams, diagnosing, and determining the best treatments** (see Figure 1). To achieve this, we conducted an experimental study with medical students, who were tasked with solving two clinical case studies in an emergency room setting, with varying levels of complexity of the case. This setting was chosen because it provides a realistic environment in which all three key medical decisions must be made. The task descriptions are as follows:

Resolution of two emergency room clinical cases with different levels of difficulty.

Digital clinical case studies are an effective method for simulating real-life scenarios, encompassing the three critical decision points while providing measurable data on actions and performance [34]. By integrating AI systems into these cases, we can assess how effectively AI supports each decision point and gain insights into its overall impact on medical practice. We collaborated with *UpHill*, a Portuguese startup specializing in healthcare software, which offers a digital, gamified tool for simulating clinical cases studies. The case studies begin with gathering patient details and performing a physical examination. As these were digital clinical cases, these tasks were simulated virtually: patient details were displayed in an initial window (see Figure 6a in the appendix), and the physical examination was conducted by selecting specific examination options, after which the corresponding information appeared automatically. Following this initial assessment, diagnostic exams (e.g. imaging, lab tests, etc.) could be prescribed, and their results became available shortly thereafter. After receiving the exam results, a primary diagnosis is made, followed by a treatment plan tailored to the patient and the specific disease. The platform features an interface that facilitates exam prescription, diagnosis, and treatment planning. Additionally, it includes a scoring system that evaluates users' decisions by comparing them to the expected resolution and actions predefined for each case study. For a detailed description of the interface, see Section A in the appendix.

3.1.1 Clinical Cases. We selected two emergency room clinical case studies from the *UpHill* Simulate tool. Both cases were fully simulated and did not rely on any personal real patient data. To assess the impact of AI across different levels of difficulty, we selected one case to be more challenging than the other. Below, we describe both cases. See the appendix B for the full medical description, including expected actions.

Low Difficulty Case - "PTE": This case involves a female patient who presents to the emergency room with Pulmonary Thromboembolism (PTE), a condition in which a blood clot obstructs a pulmonary artery, leading to impaired blood flow [47]. The diagnosis in this case is relatively straightforward due to the clear symptoms presented. The standard diagnostic tests for such cases include an electrocardiogram, Angio-CT, chest X-ray, and specific laboratory tests such as D-dimer levels [47]. The recommended treatment plan typically involves the administration of anticoagulant drugs, such as heparin. The clarity of the diagnostic phase contributes to the lower difficulty level of this clinical case. Hereafter, this case will be referred to as **PTE**.

Medium Difficulty Case - "PAC": This case involves a male patient who presents to the emergency room with Sepsis, a severe, life-threatening response to infection that triggers systemic inflammation and organ dysfunction [22]. Sepsis is a complex condition, often originating from infections such as pneumonia, and

requires a thorough assessment of multiple symptoms. Key diagnostic evaluations include a comprehensive laboratory workup, such as a Complete Blood Count (CBC), chemistry panel, and biomarkers like C-reactive protein (CRP) or lactate [22]. The treatment plan varies depending on the source and severity of the infection, with early antimicrobial therapy being a crucial and highly effective intervention. Diagnosing sepsis is particularly difficult for less experienced doctors because its symptoms often overlap with those of other conditions like pneumonia, making it hard to determine whether a patient has a localized infection or a more severe systemic condition. Hereafter, this case will be referred to as **PAC**.

3.2 Hypotheses

Our research questions aimed to assess AI's impact on diagnostic accuracy and exam prescription (**RQ.1**) and the overall efficiency of medical processes in terms of time and resource use (**RQ.2**). Additionally, we questioned whether AI support has a greater impact in complex cases or simpler ones (**RQ.3**). These questions formed the foundation for our hypotheses. This study aims to extend and validate prior findings by situating AI recommendations and explanations within a *holistic* perspective that captures the sequential and multi-staged nature of clinical decision-making. While our hypotheses are grounded in earlier research, evaluating AI support across multiple sequential decision points in a realistic workflow may reveal new patterns and trade-offs.

Main Hypothesis: AI recommendations and explanations enhance doctors' performance in solving clinical cases.

Our literature review highlights the significant potential of AI recommendations and explanations to enhance medical decision-making. Research indicates that AI can improve diagnostic accuracy and treatment planning, often surpassing human performance by leveraging vast data resources [31]. Explanations further increase doctors' trust in AI, facilitating system integration and improving transparency [40]. Despite some research, such as that by Cabitza [13], suggesting that the contribution of XAI techniques could be null in healthcare, we infer that providing robust, textual explanations for AI predictions will enable doctors to make more accurate diagnoses compared to using AI recommendations alone [17]. Based on this, we formulate the following hypotheses:

Hypothesis 1.1 (H1.1): AI recommendations improve medical diagnosis accuracy.

Hypothesis 1.2 (H1.2): The presence of explanations improves medical diagnosis accuracy compared to using only AI recommendations.

Hypothesis 2 (H2): With the assistance of AI recommendations and explanations doctors prescribe more appropriate and necessary exams.

Hypothesis 3 (H3): With the assistance of AI recommendations and explanations, doctors spend less time solving clinical cases compared to without any assistance.

Hypothesis 4 (H4): The effectiveness of AI recommendations and explanations is higher in the more difficult clinical case compared to the easier case.

3.3 Experimental Design

To test our hypothesis, we manipulated the AI assistance level in the clinical assessment task:

Control: Resolution of the two clinical cases studies without AI assistance, serving as the control group.

AI: Resolution of the two clinical cases studies receiving AI recommendations on different decision points of the clinical cases.

XAI: Resolution of the two clinical case studies receiving both AI recommendations and AI explanations on different decision points of the clinical cases.

The study used a between-subjects design for the three conditions with a counterbalancing technique which randomises the order of the two clinical case studies presented, as a way to deal with order effects.

3.3.1 The AI Assistant. We developed a scripted AI agent called *Dra. Aida*. This agent was designed to provide AI-assisted support in clinical cases by offering AI recommendations and explanations for the two experimental conditions involving AI assistance. Rather than developing new AI and explainable AI algorithms from scratch, our goal was to investigate how AI assistance impacts clinicians' performance in a realistic setting without the added complexity of building such algorithms. The AI agent delivers recommendations at two key decision points:

Exam Recommendation: The AI agent suggests three relevant exams based on the patient's information to aid in accurate diagnosis.

Diagnosis Recommendation: For each exam prescribed the AI agent suggest the most probable diagnosis based only on that exam result and available information about the patient.

Figure 2 shows the AI integration in both conditions.

Treatment planning was intentionally excluded from the experimental scope. Our participants were fifth-year medical students in their general medical training, who had not yet entered speciality training. While they can realistically simulate early-stage physicians in diagnostic reasoning and in proposing appropriate examinations or tests, they lack the clinical experience required for treatment planning. Including AI assistance at this stage would therefore not have provided a valid simulation of real-world medical decision-making and could have introduced artificial overreliance on AI recommendations. Nevertheless, even though the AI agent does not intervene in all decision points, our analysis remains *holistic*, as it assesses the impact of AI recommendations in multiple key decision points within a realistic clinical case assessment setting.

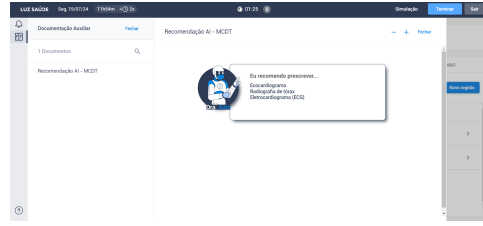
AI Assistant development. Given recent advancements in Large Language Models (LLMs) for explainable AI (XAI), leveraging GPT-4 for this task aligns with prior research that explores LLMs as tools for enhancing interpretability and user comprehension. As highlighted in the literature, LLMs can improve explanation quality by generating human-like narratives that are more accessible than traditional XAI methods. Therefore, we used GPT-4 to generate and refine recommendations and explanations for the two experimental conditions that included AI assistance.

For exam recommendations, we provided GPT-4 with patient data and prompted it to suggest and rank the most relevant exams and diagnostic tests for each case. Note that the AI recommendations were not exhaustive; only the most relevant exams were suggested, while additional exams were still expected to be ordered. For diagnosis recommendations, we prompted GPT-4 with patient history and exam results, generating potential diagnoses. It is important to note that the AI generated each diagnosis separately for each exam result, without accumulating knowledge from previous results. The recommendations and explanations were produced simultaneously in a single step using a structured prompt provided to GPT-4. From the generated responses, we extracted the most relevant recommendations along with their corresponding explanations. These outputs were then reviewed and curated for content accuracy by an expert clinician.

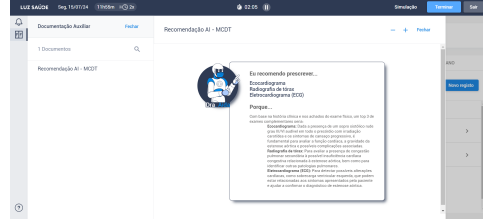
3.3.2 Clinical Case AI Resolution Paths. For each clinical case, we briefly summarize the AI agent's recommendations. For a complete description of the expected actions required to solve the cases independently of AI assistance, including exam prescriptions and diagnoses, see Section B in the appendix.

PTE Case Resolution Paths: This case presents a relatively straightforward diagnosis of PTE. The AI recommendations focus on specific exams such as chest X-ray, electrocardiogram, and laboratory tests (Troponin I, D-dimers, NT-proBNP detailed in explanation for **XAI** condition). Importantly, the critical exam, Angio-CT, which is typically essential for diagnosing PTE, was omitted from the AI's exams recommendation². This exclusion was intentional to observe whether students would independently prescribe it or rely solely on the AI. Instead, we recommended a chest X-ray, which, while not directly useful for PTE, might help in differential diagnosis.

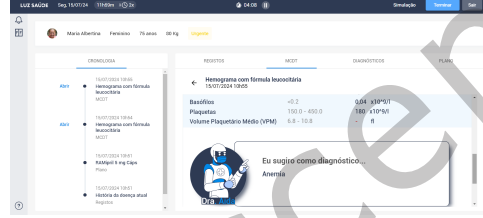
²During curation of the GPT-4 responses, this exam was deliberately excluded from the AI-generated exam's recommendations.



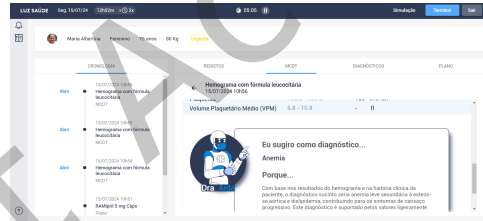
(a) Exams Recommendation in AI condition



(b) Exams Rec. and Explanation in XAI condition



(c) Diagnosis Recommendation in AI condition



(d) Diagnosis Recommendations in XAI condition

Fig. 2. Uphill Interface of Exams and Diagnosis Recommendations and Explanations

The AI agent correctly diagnosed PTE based on two key exams: Angio-CT and D-dimers. However, in the other exams that were expected to be ordered for this case, AI diagnoses varied, including suggestions such as pulmonary nodule, right bundle branch block, anemia, and myocardial infarction. These diagnoses were based on individual exam results, and they did not capture the full clinical picture and were intentionally put that way to study the diagnoses patterns. Figure 3 summarise the AI suggestions in this case within the clinical case resolution path and based on the correct actions in each decision point.

PAC Case Resolution Paths: This case focuses on diagnosing sepsis. AI recommended three exams: arterial blood gas, chest X-ray, and laboratory tests (Complete Blood Count, C-reactive protein, and creatinine levels detailed in explanation for XAI condition). For this case study, a key exam to diagnosing sepsis was the arterial

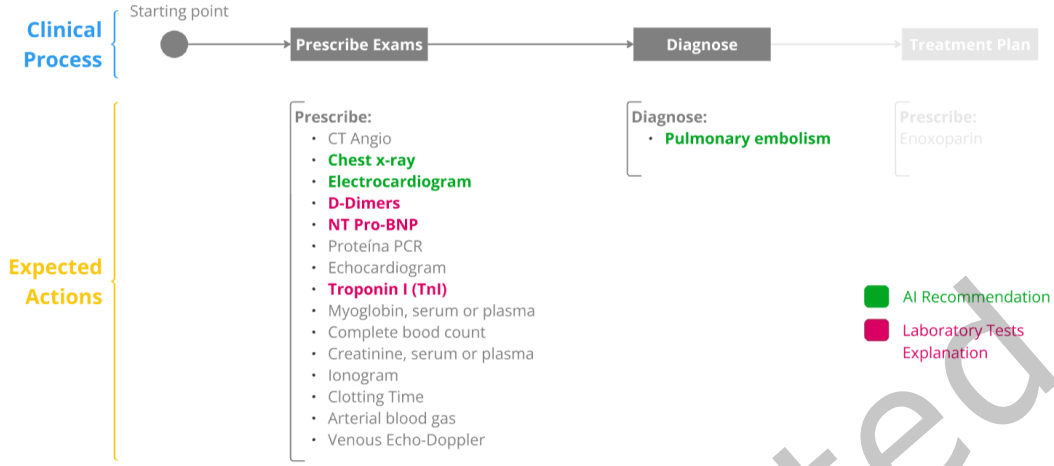


Fig. 3. PTE Case Resolution with AI Paths

blood gas exam, which revealed elevated lactate levels (19 mmol/L, where normal is below 2.5 mmol/L), a critical indicator of sepsis. While AI pointed students towards the correct diagnosis of sepsis based on this exam, other exams mostly diagnose pneumonia rather than sepsis. Diagnosing pneumonia is not incorrect, but failing to timely identify sepsis could be critically dangerous, as sepsis requires urgent intervention. Figure 4 summarise the AI suggestions in this case within the clinical case resolution path and based on the correct actions in each decision point.

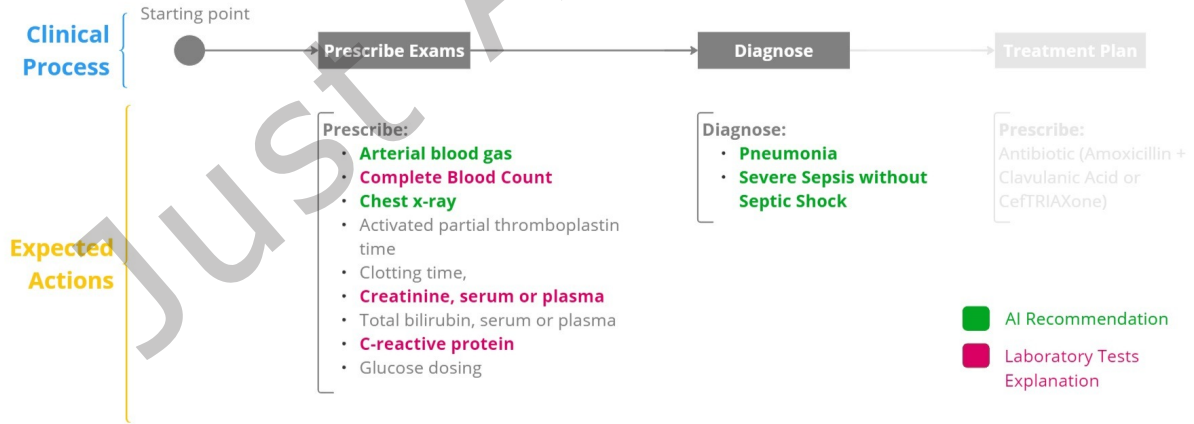


Fig. 4. PAC Case Resolution with AI Paths

3.4 Measures

3.4.1 Behavioural Measures. We developed behavioural metrics to evaluate AI's impact on clinical case resolution performance, efficiency and reliance on AI, following the clinical resolution path (see Figure 5). Each metric was calculated separately for each clinical case.

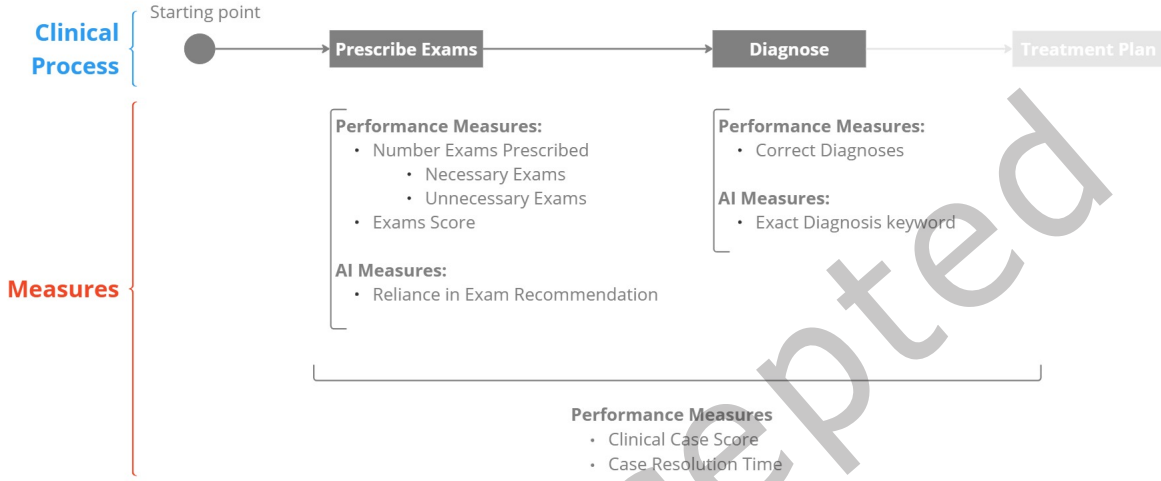


Fig. 5. Behavioural Measures Based on the Clinical Process.

Number of Exams Prescribed: Total number of prescribed exams. In a clinical case, the number of prescribed exams should balance certainty and efficiency. Clinicians should aim to order exams that provide the clearest path to diagnosis while also minimizing unnecessary tests. This is why we also tracked the **number of necessary exams**, which refers to the essential exams based on UpHill's expected actions (see Appendix B for details on expected actions for each clinical case). A higher number of necessary exams contributes positively to the case resolution score. Additionally, we recorded the **number of unnecessary exams**, calculated as the difference between the total prescribed exams and the necessary exams. A lower number of unnecessary exams improves the case resolution efficiency.

Clinical Case Score and Exams Score: The UpHill Simulator provides a list of expected actions for each clinical case by assigning points to actions based on their importance. Since the AI agent only makes recommendations for exam and diagnosis decision points, we calculate the **Clinical Case Score** as the sum of points from exam prescriptions and diagnosis actions. We also compute an **Exam Score** by summing all the points assigned to prescribed exams. We then normalize both measures by dividing them by the total possible points for a perfect clinical case resolution.

Note that the difference between the **Exam Score** and the **Number of Necessary Exams** lies in the weighted sum of each prescribed exam. While multiple exams may be necessary, some are more important than others. This subtle distinction is captured in the **Exam Score** and **Clinical Case Score** metrics. Higher Clinical Cases and Exam Scores mean better clinical case resolution.

Reliance in Exam Recommendation: We assess whether AI recommendations were followed during the exam prescription phase by calculating the number of exams prescribed by users that were also

recommended by the AI, normalized by the total number of exams suggested by the AI assistant. In the **Control** group, where no AI assistance is provided, reliance cannot be measured. However, this group serves as a baseline for comparison with the other two conditions. Higher rates of reliance meant that a great proportion of AI suggestions were followed.

We also tracked whether specific exam prescription suggestions were followed. In the *PTE* case, we monitored **Chest X-ray**, as this exam is unnecessary if an Angio-CT is performed, yet it was suggested by the AI in the conditions with AI assistance. Similarly, in the *SEP* case, we tracked **Arterial Blood Gas testing**, as it is the only exam where the AI assistant recommends a sepsis diagnose and was recommended by the AI.

Correct Diagnosis: We tracked if the correct diagnoses for each clinical case was made. Multiple terms may qualify as correct diagnoses, for example, “Pulmonary Embolism” can also be diagnosed using the name “Embolia pulmonar sem cor pulmonale agudo” or “Severe Sepsis Without Septic Shock” can be diagnosed using the name “Sepsis, unspecified organism”, according to the ICD-10-CM³. Therefore we created a binary variable that indicates if the diagnosis was correct based on a list of acceptable diagnoses⁴.

Exact Diagnosis Keywords: We assessed whether AI recommendations were followed at the diagnosis decision point by tracking cases where the diagnosis matched the AI’s suggestion exactly. To do this, we created a binary variable that equals 1 if the diagnosis made was identical to the AI-recommended diagnosis. In the *PTE* case, the keyword was “Pulmonary Embolism”, and in the *PAC* case, it was “Severe Sepsis Without Septic Shock”. Since multiple terms could qualify as a correct diagnosis, there was a possibility that a diagnosis was correct but did not match the AI’s suggestion word for word. In such cases, we considered the diagnosis correct but did not count it as reliance on the AI at the diagnosis decision point.

Case Resolution Time: We measure the resolution time for each clinical case study, with shorter times indicating greater efficiency.

3.4.2 Non Verbal Measures.

Eye Tracking: We used an eye tracker to analyse gaze patterns during case resolutions, focusing on participants’ interactions with explanations and recommendations. Specifically, we tracked fixation: the most common type of eye movement, which provide insights into cognitive processes and attention. A fixation occurs when the eyes momentarily stop scanning and focus the foveal area on a specific point [45]. To classify eye movements, we applied the *I-VT algorithm*, which categorizes movements based on their velocity [43, 49]. We examined key metrics, including the total number of fixations, and total fixation time, to better understand gaze behaviour and cognitive processing.

Eye-tracking data was acquired using a **Tobii Pro Nano** (Tobii Pro AB) screen-based eye tracker [1], which records gaze data at a sampling frequency of 60 Hz. The system has a reported accuracy of 0.3° and a spatial precision of 0.10° RMS (Root Mean Square) under optimal conditions. Calibration was conducted manually by the researcher every day before the start of sessions using the Tobii Pro Eye Tracker Manager software.

Mouse Tracking: In addition to eye tracking, we recorded mouse movements to better understand engagement with the exam recommendations. Specifically, we measured the total number of clicks on recommendations⁵, and the time spent reading the recommendations.

³ICD-10-CM is a system used by healthcare providers to classify and code all diagnoses

⁴The list of acceptable diagnoses for each case was verified by a clinician

⁵Participants had to click a button to access the AI recommendations

3.5 Procedure and Pilot Tests

Participants completed the task in person at Hospital da Luz, Lisbon or the Faculty of Medicine at the University of Lisbon. Each participant was seated alone in a room at a desk with a computer, and an eye tracker was positioned at the base of the laptop to monitor gaze behaviour during the experiment. Before starting, participants signed a consent form⁶ and watched a video introducing the platform used for the simulated clinical cases. They then solved the two clinical cases in randomized order and completed a questionnaire to collect demographics data. As compensation, each participant received a cinema ticket voucher worth €6.50. To validate the study setup, we conducted four pilot tests with a recent medical graduate, two nurses, and a general practitioner. These tests provided insights into task difficulty and identified potential issues, leading to revisions and confirming the decision not to include AI recommendations for treatment plans, as medical students are not trained for this decision. Overall, the pilot feedback was positive, requiring no major changes to the study design.

3.6 Participants

Given the demanding schedules of physicians, we chose to recruit senior medical students, as they have significant experience in solving clinical cases. This decision ensured a realistic simulation environment while allowing us to effectively capture and analyse decision-making behaviour across all key decision points. Participants were required to be in their fifth year or higher, fluent in Portuguese (since clinical cases were presented in Portuguese), and at least 18 years old. We recruited students from the Faculty of Medicine at the University of Lisbon.

Participants were randomly assigned to one of three experimental conditions: Control, AI, or XAI. After the initial data collection, we observed that a subset of participants in the AI and XAI conditions did not view the AI exam recommendations, as confirmed by screen and mouse recordings. To ensure balanced data across conditions and sufficient exposure to the AI manipulation, additional participants were subsequently recruited for the AI and XAI conditions. Following data verification, a total of 44 medical students participated in the study but 10 participants were excluded because they failed to view the AI recommendations, resulting in 34 valid task completions (Control: $n = 11$; AI: $n = 12$; XAI: $n = 11$). Among the 34 participants with valid task completions, 66% identified as female and 34% as male. Regarding age distribution, 87% were between 18 and 25 years old, 11% were between 25 and 35 years old, and 2% (one participant) was between 35 and 50 years old. The majority of participants (48%) were in their sixth year of medical school, 32% were in their fifth year, and 20% were general training interns⁷.

4 Results

We conducted all statistical analyses using IBM SPSS Statistics version 28. To evaluate differences across AI assistance levels (**Control, AI, and XAI**), we performed one-way ANOVA tests. When significant differences were identified, we conducted Tukey post-hoc tests for pairwise comparisons. If no significant differences emerged, we combined the AI and XAI groups and employed planned contrasts to determine whether AI assistance in any form (**AI + XAI**) differed from having no AI assistance (**Control**). For binary outcome variables related to diagnosis or specific exam prescriptions, we utilized Chi-Square tests to assess group differences. All ANOVA assumptions were verified prior to analysis. When normality or homogeneity assumptions were not met, equivalent non-parametric tests were applied, as indicated in the corresponding result descriptions. Descriptive results are reported as means and standard errors ($M \pm SE$), while global summary values are presented using standard deviations (SD).

A repeated measures ANOVA was not employed because our analysis did not aim to directly compare the two clinical cases (*PTE* and *PAC*). Given that these cases involve inherently different clinical reasoning processes and resolution paths, treating them as repeated measures would not be statistically appropriate. Instead, our

⁶The consent form and the user study was reviewed and approved by the Data Protection Officer of our research Institute, INESC-ID

⁷General training interns are medical graduates in their first year of postgraduate clinical training

goal was to evaluate the effect of AI assistance independently within each case, justifying the use of separate one-way ANOVAs for each condition. Only significantly ($\alpha < 0.05$) results are reported in this section. For a full description of the statistical analyses see appendix C.

4.1 Behavioural Measures

4.1.1 Number of Exams Prescribed. In the **PTE** case the average number of exams prescribed was 6.38 ($SD = 2.92$). The number of exams prescribed are similar between conditions with no significance differences found ($F(2, 31) = .085, p = .918$). However, in the most challenging scenario, **PAC**, the control group (**Control**) shows a lower average number of prescribed exams ($5.91 \pm .90$) compared to the other conditions (8.42 ± 1.52 for **AI** and 8.27 ± 1.69 for **XAI**). However, the difference was not statistically significant ($F(2, 31) = .974, p = .389$).

4.1.2 Number of Necessary Exams. In both clinical cases, the mean values across the different conditions were very similar. For the **PTE** case the average number of necessary prescribed exams was 4.56 ($SD = 1.93$). For the **PAC** case, the average number of necessary exams was 3.47 ($SD = 1.29$). The ANOVA tests did not reveal statistically significant differences between the conditions for either clinical case.

4.1.3 Number of Unnecessary Exams. In the **PTE** case the average number of unnecessary exams prescribed was 1.82 ($SD = 1.90$). The number of unnecessary examinations prescribed is similar between conditions with no significant differences found ($F(2, 31) = .171, p = .844$). However, in the **PAC** clinical case, the control group (**Control**) shows a lower average number of prescribed exams that were unnecessary ($2.55 \pm .77$) compared to the other conditions (4.75 ± 1.72 for **AI** and 4.91 ± 1.97 for **XAI**). However, the difference was not statistically significant ($F(2, 31) = .684, p = .512$).

4.1.4 Exams Score. In the easiest case, **PTE**, the **AI** and **XAI** conditions have higher mean exams scores ($.457 \pm .031$ and $.408 \pm .052$ respectively) when compared to the **Control** group ($.321 \pm .052$). The ANOVA test did not reveal a statistically significant difference between the three conditions ($F(2, 31) = 3.135, p = .058$). However, conducting planned contrast tests to compare the **Control** group with the **AI + XAI** group, we obtained a statistically significant result ($F(2, 31) = 23.12, p = .028$), indicating significant differences between the Control group and the groups with AI assistance. Regarding the difficult case, **PAC**, the average exam score was 0.262 ($SD = .059$) and the values are very similar between conditions. We did not find statistical significances between the three conditions.

4.1.5 Clinical Case Score. Clinical case scores were higher with AI assistance in both cases. The statistical results are presented in Table 1. In the **PTE** case, the difference was not statistically significant ($F(2, 31) = 3.108, p = .059$). However, planned contrasts further revealed a statistically significant difference between the **Control** group and the **AI + XAI** group ($F = 2.462, p = .02$). In the more complex **PAC** case, significant differences emerged between conditions, with the **XAI** group performing significantly better than the **Control** group ($p = .015$).⁸

4.1.6 Reliance in Exam Recommendation. In the **PTE** case, participants in the **XAI** group exhibited the highest reliance on AI suggestions (0.509 ± 0.073), followed by the **AI** group (0.417 ± 0.057), with the **Control** group serving as the baseline (0.309 ± 0.073). However, these differences in AI reliance were not statistically significant. In the more complex **PAC** case, reliance differences were even smaller, with no statistical evidence of variation across conditions ($F(2, 31) = 0.363, p = .699$).

When analysing reliance on the chest X-ray prescription, that was suggested by the AI system but optional for the resolution path, no significant association was observed between prescription rates and conditions ($p = .061$), nonetheless, the results suggest a trend where participants with AI assistance were more likely to prescribe a chest X-ray than those without AI support. Similarly, for arterial blood gas prescription, no significant association

⁸Although the data did not meet the assumption of homogeneity of variances, a non-parametric test confirmed this result.

	PTE		PAC	
	Mean	SE	Mean	SE
Control	0.313	0.041	0.273	0.030
AI	0.450	0.039	0.313	0.012
XAI	0.431	0.046	0.369	0.023
F(2, 31)	3.108		4.517	
p-value	.059		.019*	

Table 1. Mean and Standard Error of Clinical Case Score and p-values for ANOVA tests.

was found between prescription rates and condition ($p = .081$). However, all participants in the **AI** condition prescribed it, compared to 91% in the **XAI** condition and 73% in the **Control** group. Tables 2 and 3 present the results of these two particular exams.

	Chest x-ray Prescription (%)
Control	27.3
AI	66.7
XAI	72.7
p-value	0.061

Table 2. Percentage of Chest x-ray Prescription in **PTE** Case by Condition and $\chi^2(2)$ p-value

	Arterial Blood Gas Prescription (%)
Control	72.7
AI	100
XAI	90.9
p-value	0.081

Table 3. Percentage of Arterial Blood Gas Prescription in **PAC** Case by Condition and $\chi^2(2)$ P-value

4.1.7 Correct Diagnosis. Table 4 shows the percentage of correct diagnosis by each of the three conditions (**Control**, **AI** and **XAI**) across the three main diagnoses: **Pulmonary Embolism** for the easier case, **PTE**, and **Pneumonia** and **Sepsis** for the **PAC** case, along with the p-values for the χ^2 tests. In all the three diagnoses **XAI** was the group that have higher levels of diagnosis accuracy, followed by the **AI** condition.

In the **PTE** case, when diagnosing pulmonary embolism, we did not find a statistically significant association between pulmonary embolism diagnosis and the AI assistance level ($p = 0.098$). However, the **XAI** condition had a higher average rate of correct diagnoses compared to the other conditions. In the more complex case, **PAC**, there were no statistically significant differences between conditions for diagnose of Pneumonia - most of the medical students correctly diagnosed pneumonia. Nonetheless, for the diagnose of Sepsis, the **XAI** condition had a higher rate of sepsis diagnoses compared to the other conditions, although this difference was not statistically significant ($p = 0.086$).

4.1.8 Exact Diagnosis Keyword. Table 5 shows the percentage of diagnosis that followed the exact wording as the AI recommendation by each of the three conditions (**Control**, **AI** and **XAI**) across the two main diagnoses: **Pulmonary Embolism** for the easier case, **PTE**, and **Sepsis** for the **PAC** case, along with the p-values for the χ^2 tests. Similarly to the Correct diagnosis measure, in both diagnoses **XAI** was the group that had higher levels of exact diagnosis keyword.

Despite participants in the **XAI** condition diagnosed “Pulmonary Embolism” more frequently than **AI** and specially control group, no significant association was found between the diagnosis and the AI assistance level. However, in the more complex case involving the exact diagnosis term “Severe Sepsis Without Septic Shock”, there were statistically significant differences between the conditions, particularly when comparing the **XAI** condition (45.5%) with the **Control** (9.1%) and **AI** (0.0%) conditions.

	Pulmonary Em- bolism (%) - PTE case	Pneumonia (%) - PAC case	Sepsis (%) - PAC case
Control	54.5	90.9	18.2
AI	58.3	100	41.7
XAI	90.9	100	63.6
p-value	.098	.313	.086

Table 4. Percentage of Correct Diagnosis by Condition and $\chi^2(2)$ p-value

	Pulmonary Em- bolism (%) - PTE case	Sepsis (%) - PAC case
Control	27.3	9.1
AI	41.7	0.0
XAI	54.5	45.5
p-value	0.423	0.007**

Table 5. Percentage of Exact Diagnosis Keyword by Condition and $\chi^2(2)$ p-value

4.1.9 Case Resolution Time. Regarding the time to solve clinical cases, participants with access to AI assistance, on average, spent less time on case resolution, the time being particularly lower in the **XAI** condition. In the **PTE** case, the time (in minutes) to resolve the case was (20.07 ± 2.242) for the **Control** group; (15.34 ± 1.514) for the **AI** group and (13.97 ± 1.330) for the **XAI** group. The ANOVA test showed that there was a statistically significant difference between the conditions ($F(2, 31) = 3.350 = .048$) and the post hoc test indicated that the differences were between the **Control** and the **XAI** conditions. In the **PAC** clinical case, the differences were not statistically significant ($F(2, 31) = .803, p = .457$).

4.2 Non Verbal Measures

4.2.1 Eye Tracker. Four records were discarded due to poor calibration, resulting in 30 valid records (10 in each condition).

Table 6 summarizes the number of fixations and the total fixation duration across the three experimental conditions. In both clinical cases, total fixation duration decreased with the introduction of the **AI** and **XAI** systems compared to the **Control** condition. The one-way ANOVA conducted on the total fixation duration for the **PTE** clinical case revealed a statistically significant effect ($F(2, 27) = 5.842, p = .008$). Post hoc comparisons using the Bonferroni correction indicated that fixation durations were significantly lower in both the **AI** ($p = .01$) and **XAI** ($p = .024$) conditions compared to the **Control**. However, no statistically significant differences were observed in the number of fixations across conditions, as indicated by the corresponding ANOVA results. Regarding the more complex case, **PAC**, no statistically significant differences were found in either total fixation duration or number of fixations.

4.2.2 Mouse Tracker. Table 7 presents the total number of clicks on AI recommendations and the duration of visual engagement with these recommendations across the two clinical cases and the AI-assisted conditions. Participants in the **XAI** condition spent significantly more time viewing the recommendations than those in the **AI** condition, regardless of case difficulty. Regarding the number of clicks, no significant difference was observed

	PTE		PAC	
	Number of Fixations	Total Fixation Duration (s)	Number of Fixations	Total Fixation Duration (s)
Control	691.56	1013.927	656.56	1004.184
AI	549.60	629.783	548.40	696.719
XAI	535.45	680.678	668.27	786.668
F(2,27)	1.298	5.842	0.402	1.387
p-value	0.290	0.008**	0.673	0.267

Table 6. Fixation Metrics and p-values for ANOVA tests

between conditions in the simpler case (**PTE**). However, in the more complex case, **PAC**, participants in the **XAI** condition clicked significantly more often than those in the **AI** condition, a difference that was statistically significant ($F(2, 21) = 10.470$, $p = .004$).

	PTE		PAC	
	Number of Clicks	Visualisation Duration (s)	Number of Clicks	Visualisation Duration (s)
AI	1.917 $\pm$ 1.379	6.047 $\pm$ 2.484	.833 $\pm$.577	4.406 $\pm$ 2.996
XAI	1.818 $\pm$ 1.079	14.127 $\pm$ 7.721	2.272 $\pm$ 1.421	14.089 $\pm$ 7.767
F(2,21)	.036	11.850	10.470	16.086
p-value	.852	.002**	.004**	<.001***

Table 7. Average Number of Click and Visualisation Duration ($\pm SE$) and ANOVA results

5 Discussion

In this study, we investigate the impact of AI assistance on the resolution of clinical cases, specifically when AI suggests decisions, sometimes accompanied by explanations, throughout a full process of a clinical case, including exam prescriptions and clinical diagnoses. Our primary objective is to understand how an AI assistant influences clinical performance in exam prescriptions and clinical diagnosis (**RQ.1**), how clinical efficiency is affected in these settings (**RQ.2**), and how AI assistance interacts with different levels of case complexity (**RQ.3**).

We hypothesized that both AI recommendations (**H1.1**) and explanations (**H1.2**) will enhance medical performance in clinical diagnosis. Additionally, we expected that AI assistance would lead to more appropriate and necessary exam prescriptions (**H2**). Furthermore, we hypothesized that AI-assisted clinicians would exhibit greater efficiency, spending less time solving clinical cases (**H3**), and that the effectiveness of AI assistance would be more pronounced in more challenging clinical cases (**H4**).

To test these hypotheses, we conducted a user experiment with final-stage medical students, requiring them to solve two clinical cases of varying difficulty while tracking key measures of performance, efficiency, and reliance on AI. In the following sections, we discuss our results, their alignment with previous research, and their broader implications for clinical decision-making.

5.1 AI Recommendations and Explanations Can Enhance Medical Performance Without Guaranteeing Efficiency

We evaluated medical accuracy by analysing two main indicators: the appropriateness of exam prescriptions and the correctness of the final diagnoses. In the easier clinical case, medical students performed better in exam prescription when supported by AI, both in the standard AI condition and the explainable AI condition. In the more complex clinical scenario, participants with AI support were better to correctly diagnose sepsis, a particularly challenging task for inexperienced clinicians. Overall, medical students achieved higher clinical scores (which combine exam prescription accuracy and diagnostic accuracy) when they received AI recommendations, with or without explanations. These results support our first hypothesis that AI can enhance medical performance (H1.1). When it comes to the effect of explanations specifically, we observed that in the more difficult case, providing explanations helped participants make a correct diagnosis of sepsis, thereby boosting their overall performance. However, explanations alone did not lead to noticeable improvements in exam prescription. Thus, our second hypothesis, that explanations can independently improve medical accuracy, was only partially supported (H1.2). The improvement in performance was most evident when combining both AI and XAI conditions, especially compared to the control group. This suggests that the impact of explanations may depend on the difficulty of the case. A similar conclusion was drawn by Cabitza et al. [13], who found that explanations often have limited effects, and that the AI's recommendation itself plays a more central role in decision-making when looking only at final medical performance.

Turning to efficiency, we found no evidence that AI assistance led to fewer or more necessary exams. On the contrary, in the difficult clinical case, students in both AI and XAI groups tended to prescribe more tests, including unnecessary ones, compared to the control group. This indicates a drop in efficiency, rather than an improvement. One plausible explanation relates to the complexity of the case. For example, nearly all participants with AI support ordered the arterial blood gas test, which was the only test pointing toward sepsis — a difficult diagnosis for novice students. We believe that the AI recommendation in this result heightened participants' awareness of the diagnostic challenge, prompting them to adopt a more cautious and deliberative approach. While this helped accuracy, it also led to over-prescription, which runs counter to our hypothesis that AI would improve exam efficiency (H2). Moreover, the time required to solve the case did not differ significantly across conditions in the more complex scenario, further suggesting that AI support does not necessarily enhance efficiency in solving complex clinical cases.

That said, AI assistance did help reduce the time needed to solve the easier clinical case. This effect is even more evident when examining the eye-tracking data. In the easier case, participants fixated on the screen for considerably longer periods when they had no AI assistance. In contrast, when they received AI suggestions, particularly those accompanied by explanations, the duration of fixations was noticeably shorter. This partially supports our third hypothesis, which proposed that AI would reduce time spent per case (H3). This finding supports the idea that AI systems of this kind can help clinicians spend less time on each case, especially more straightforward ones in which clinicians likely already have a strong initial impression of the underlying clinical problem based on patient details. In such situations, AI assistance may serve primarily to confirm existing suspicions and thereby reduce cognitive load, a factor that may help mitigate fatigue and overwork in clinical practice. The shorter fixation durations observed in the eye-tracking data further support this interpretation, as reduced visual attention demands are often associated with lower cognitive load during decision-making.

5.2 Explanations Can Increase the Reliance in the AI Recommendations

One of our goals in this study was to explore how participants relied on AI recommendations, particularly on key decision points. To do this, we tracked reliance in two areas: the prescription of specific exams, and the use of exact keywords recommended by the AI in the diagnosis.

In the simpler case, the AI recommended performing a chest X-ray. While this test could help narrow down diagnostic options, it was not sufficient to diagnose pulmonary thromboembolism (PTE) and therefore should be considered an optional exam. The results show that participants with AI assistance prescribed this exam more often than those without AI support (66.7% and 72.7% vs. only 27.3%). This highlights a tendency to rely on AI recommendations, even when the suggested exam is optional and other diagnostic tools, such as CT angiography, could provide a definitive diagnosis without it.

Focusing on the more complex scenario, the AI assistant recommended prescribing the arterial blood gas test. This was the only exam that, when reviewed, would trigger a sepsis diagnosis recommendation. Medical students in the AI-supported conditions (both **AI** and **XAI**) were more likely to prescribe this test than those in the control group (100% and 91%, compared to 72.7%, respectively). This pattern suggests that participants relied on the AI's recommendations. To further understand the extent of this reliance, we analysed whether participants used the exact diagnostic term suggested by the AI — “Severe Sepsis Without Septic Shock”, rather than a similar or approximate term. Interestingly, none of the participants in the **AI** condition used this specific phrasing, whereas 46% of those in the **XAI** condition did. This indicates a higher degree of reliance on the AI system when explanations were provided alongside recommendations. Non-verbal behaviour supports this finding as well. Participants in the **XAI** group viewed the AI's exam suggestions (that is, a higher number of clicks recorded through mouse tracking), more frequently than those in the **AI** group when solving the **PAC** case. This increased reliance appears to stem from the explanations offered in the **XAI** condition, which may have helped participants better understand the clinical scenario and the reasoning behind the AI's suggestions.

These findings are consistent with prior research, which has shown that explanations can enhance trust and reliance in AI systems [12, 44]. While reliance can be beneficial when AI recommendations are accurate, as was the case in our study, it's important to note the potential downside of relying on the explanations because they are present [20]. Over-reliance becomes problematic when the AI makes incorrect or misleading suggestions. Our study did not include such scenarios, so we were not able to assess over-reliance. However, this remains a critical direction for future research.

5.3 Case Complexity Shapes AI Impact

This study examined the resolution of two clinical cases with differing levels of complexity to evaluate how AI assistance influences diagnostic performance across varying degrees of difficulty. Results indicated that medical students benefited from AI support in both scenarios, as demonstrated by higher clinical case scores when AI assistance was available.

However, the level of case complexity shaped the effect of AI assistance. The **PTE** case was classified as relatively simple because the diagnostic path was straightforward: given the initial clinical information and appropriate tests, final-year medical students could readily identify a pulmonary embolism. In such cases, students typically formed an educated hypothesis early on and used subsequent evidence to confirm it. Eye-tracking data supported this interpretation, students with AI assistance spent less time fixating on the screen than the control group, suggesting reduced attentional demand. This likely occurred because AI recommendations aligned with their initial judgments, turning the task into a confirmatory reasoning process. Consequently, these cases were solved more rapidly. Although diagnostic accuracy was still higher with AI, the gain mainly reflected greater efficiency and confidence in confirming an already-correct hypothesis. Thus, in simpler cases, confirmatory AI recommendations can enhance both accuracy and efficiency.

In contrast, the **PAC** case represented a more complex diagnostic challenge, where identifying sepsis was non-trivial. While pneumonia was an obvious and easily recognized diagnosis, reaching the correct conclusion required integrating multiple cues and reasoning beyond initial impressions. Here, AI assistance, particularly when accompanied by explanations, helped students arrive at the correct diagnosis more often. With explanations,

participants engaged more deeply with the system, revisiting AI suggestions and checking recommended exams more frequently. This pattern indicates both increased reliance and active engagement, as the case was less intuitive. However, students with AI also prescribed more unnecessary exams, reflecting initial uncertainty and a need for reassurance when reasoning through a complex case.

These findings align with dual-process theories of reasoning [30], which distinguish between intuitive (Type 1) and analytical (Type 2) thinking. In simpler cases, AI appears to support intuitive, confirmatory reasoning, improving efficiency and confidence. In contrast, in complex cases, AI promotes more deliberate analytical engagement, aiding diagnostic accuracy but often increasing cognitive effort.

This suggests that AI assistance is particularly valuable for improving medical accuracy in complex cases that are harder to diagnose. These findings align with previous literature, which shows that AI support is most beneficial in situations where clinical reasoning is more demanding and diagnoses are less straightforward [14].

These results show that while the AI system was beneficial in both clinical cases, its advantages vary depending on case complexity. In more straightforward cases, AI is particularly helpful for saving time and improving efficiency, whereas in complex scenarios, it enhances clinical performance but does not necessarily lead to time savings.

5.4 Broader Implications for clinical practise:

In clinical settings, healthcare professionals remain sceptical about adopting AI tools that may disrupt established workflows [3, 64]. Despite significant advances in AI development and a growing body of research highlighting its potential benefits, the integration of AI into routine clinical practice remains limited due to sociotechnical challenges that are difficult to address. These challenges include integrating AI systems into existing healthcare infrastructures and designing user interfaces that may not align with clinicians' needs and expectations [58, 64].

This study, although conducted with medical students in a simulated environment, offers valuable insights into how AI may influence clinical practice. Medical students solving clinical cases can serve as reasonable proxies for early-career clinicians: they are highly motivated to achieve correct diagnoses and avoid errors. The pressure they experience to perform well mirrors the pressures faced by practicing clinicians. As such, the ways in which students interacted with AI in this context can inform our understanding of how AI might function in real-world clinical settings. Our findings suggest that AI systems have the potential to enhance medical performance, particularly when their suggestions are accompanied by explanations. The presence of explainability appears to build trust, encouraging clinicians to follow AI guidance, even when it involves prescribing optional diagnostic tests. However, AI support did not consistently lead to improved efficiency. In fact, when confronted with complex cases, participants tended to order additional tests, likely in an effort to validate or better understand the AI's recommendations. This behavior highlights a central tension in the adoption of clinical AI: while it can support decision-making, it may also increase cognitive and diagnostic workload in more complex scenarios.

In contrast, during simpler clinical cases, the AI system proved useful in reducing the time required for assessment. While diagnostic accuracy is often prioritized, gains in efficiency are equally critical, especially in today's healthcare systems, which are frequently overburdened, face limited access, and contend with rising costs [5, 42]. Even modest reductions in decision-making time for straightforward cases can improve patient throughput, reduce operational costs, and ultimately enhance access to care.

Ultimately, while the full impact of clinical AI in real-world settings remains uncertain, this study underscores that its integration will likely involve trade-offs. Gains in diagnostic accuracy may come at the expense of efficiency in complex cases, and the benefits of AI will depend significantly on how well it is incorporated into clinicians' reasoning processes [36].

6 Study Limitations and Future Work

First, although the study adopted a *holistic* approach, examining the influence of AI recommendations and explanations across exam prescriptions followed by diagnostic reasoning, it did not encompass all stages of the clinical workflow. The use of the term *holistic* here refers to the integration of multiple interrelated diagnostic steps rather than the full continuum of patient management. Future work could build on this by expanding AI integration to include subsequent processes such as treatment planning, inter-provider consultation, and longitudinal patient monitoring. Such extensions would provide a more comprehensive understanding of how AI systems interact with clinicians across the full spectrum of care.

Second, the AI agent predominantly provided accurate recommendations. While this design choice was essential to establish baseline utility and isolate the effects of explanations, it limited the exploration of how clinicians respond to incorrect or uncertain AI outputs. Understanding how users handle flawed or ambiguous recommendations is critical for evaluating trust calibration and decision robustness. Future studies should incorporate AI systems with variable accuracy and confidence levels to examine how clinicians manage uncertainty and maintain appropriate trust in real-world conditions.

Third, certain methodological choices may have influenced variability in the results. The simulated clinical environment, though enabling experimental control, lacked real-world consequences, which may have reduced cognitive and emotional engagement. In addition, ten participants were excluded because they did not view the AI's exam recommendations, as verified through screen and mouse recordings. This exclusion was necessary to preserve the integrity of the experimental manipulation; however, it also highlights a meaningful avenue for future research. In real clinical settings, users may occasionally overlook or disregard AI input. Future studies should explicitly investigate the factors leading to such omissions, whether due to workload, interface design, or trust issues, and assess their impact on decision quality.

Finally, this study recruited final-year medical students to approximate early-career clinicians, but future research should extend these findings to residents and practising physicians with varying levels of expertise. Incorporating experienced clinicians operating in ecologically valid settings would clarify how expertise moderates the influence of AI recommendations and explanations across diverse clinical contexts.

7 Conclusion

This study evaluated the impact of AI recommendations and explanations on medical decision-making in realistic clinical scenarios, aiming to understand how AI assistants influence healthcare professionals' practices across several key decision points in clinical cases. Results showed that in simpler clinical cases, both AI and XAI improved exam prescription accuracy and reduced assessment time, with AI assistance combined with explanations performing especially well. In complex cases, AI assistance, particularly with explanations, improved diagnostic outcomes, although it sometimes led to increased resource use by prescribing more unnecessary exams. Participants with access to explanations relied more on AI, which enhanced outcomes.

These findings offer meaningful guidance for the thoughtful deployment of AI and XAI systems to support clinical decision-making. They underscore both the promise of these technologies and the critical need for their responsible and context-aware integration into healthcare workflows. Although real-world implementation will require careful navigation of ethical, legal, and practical challenges, this study provides a strong foundation for future research grounded in realistic clinical contexts and adopting a *holistic* analytical approach.

To our knowledge, this study is among the first to examine AI-assisted medical decision-making across multiple interdependent steps within a realistic clinical workflow. By simulating the full decision process, from exam prescription to diagnosis, within an emergency care context, it moves beyond the task-level evaluations that have dominated prior research. This broader perspective underscores the importance of studying AI support as

part of an integrated clinical process, providing a foundation for future work on decision support systems that more accurately reflect the complexity of real-world medical practice.

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References

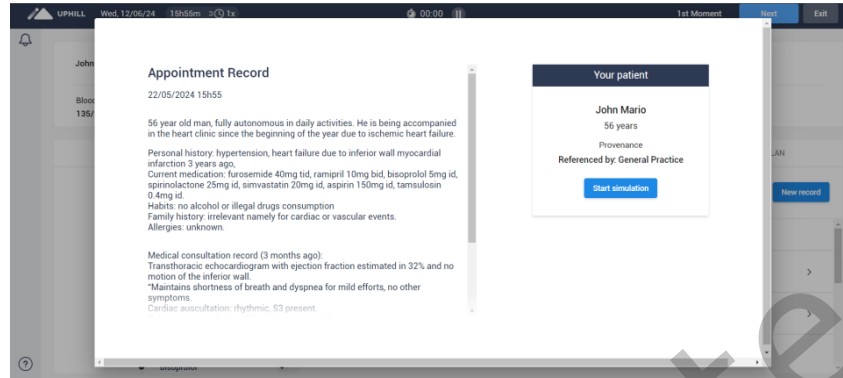
- [1] Tobii AB. 2024. Tobii Pro Lab. Computer software. <http://www.tobii.com/>
- [2] Sajid Ali, Tamer Abuhmed, Shaker El-sappagh, and Khan Muhammad. 2023. Explainable Artificial Intelligence (XAI): What we know and what is left to attain Trustworthy Artificial Intelligence. *Information Fusion* 99, March (2023), 101805. doi:10.1016/j.inffus.2023.101805
- [3] Nora Arvai, Gellért Katonai, and Bertalan Mesko. 2025. Health care professionals' concerns about medical AI and psychological barriers and strategies for successful implementation: scoping review. *Journal of Medical Internet Research* 27 (2025), e66986.
- [4] Junaid Bajwa, Usman Munir, Aditya Nori, and Bryan Williams. 2021. Artificial intelligence in healthcare: transforming the practice of medicine. *Future healthcare journal* 8, 2 (2021), e188–e194.
- [5] Roland Bal and Iris Wallenburg. 2022. Linking Costs and Quality in Healthcare: Towards Sustainable Healthcare Systems: Comment on "Hospitals Bending the Cost Curve With Increased Quality: A Scoping Review Into Integrated Hospital Strategies". *International journal of health policy and management* 12 (2022), 7461.
- [6] Gagan Bansal, Besmira Nushi, Ece Kamar, Eric Horvitz, and Daniel S. Weld. 2021. Is the Most Accurate AI the Best Teammate? Optimizing AI for Teamwork. *35th AAAI Conference on Artificial Intelligence, AAAI 2021* 13A (2021), 11405–11414. doi:10.1609/aaai.v35i13.17359 arXiv:2004.13102
- [7] Gagan Bansal, Besmira Nushi, Ece Kamar, Daniel S. Weld, Walter S. Lasecki, and Eric Horvitz. 2019. Updates in Human-AI Teams: Understanding and Addressing the Performance/Compatibility Tradeoff. *Proceedings of the AAAI Conference on Artificial Intelligence* 33, 01 (Jul. 2019), 2429–2437. doi:10.1609/aaai.v33i01.33012429
- [8] Lasse Benzing, Frank Ursin, Wolf-Tilo Balke, Tim Kacprowski, and Sabine Salloch. 2023. Should Artificial Intelligence be used to support clinical ethical decision-making? A systematic review of reasons. *BMC medical ethics* 24, 1 (2023), 48.
- [9] Astrid Bertrand, Rafik Belloum, James R. Eagan, and Winston Maxwell. 2022. How Cognitive Biases Affect XAI-assisted Decision-making: A Systematic Review. In *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society (Oxford, United Kingdom) (AIES '22)*. Association for Computing Machinery, New York, NY, USA, 78–91. doi:10.1145/3514094.3534164
- [10] Al Amin Biswas. 2024. A comprehensive review of explainable AI for disease diagnosis. *Array* 22, April (2024), 100345. doi:10.1016/j.array.2024.100345
- [11] Marvin Braun, Maike Greve, Johannes Riquel, Alfred Benedikt Brendel, and Lutz M Kolbe. 2022. Meet Your New Colle (AI) gue: Exploring the Impact of Human-AI Interaction Designs on User Performance. 1708 pages.
- [12] Adrian Bussone, Simone Stumpf, and Dymrna O'Sullivan. 2015. The Role of Explanations on Trust and Reliance in Clinical Decision Support Systems. In *2015 International Conference on Healthcare Informatics (ICHI)*. IEEE, Piscataway, NJ, USA, 160–169. doi:10.1109/ICHI.2015.26
- [13] Federico Cabitza, Andrea Campagner, Luca Ronzio, Matteo Cameli, Giulia Elena, Maria Concetta, Luca Maria, Duarte Folgado, Marília Barandas, and Hugo Gamboa. 2023. Rams , hounds and white boxes : Investigating human – AI collaboration protocols in medical diagnosis. *Artificial Intelligence In Medicine* 138, January (2023), 102506. doi:10.1016/j.artmed.2023.102506
- [14] Federico Cabitza, Lorenzo Famiglini, Caterina Fregosi, Samuele Pe, Enea Parimbelli, Giovanni Andrea La Maida, and Enrico Gallazzi. 2025. From Oracular to Judicial: Enhancing Clinical Decision Making through Contrasting Explanations and a Novel Interaction Protocol. In *Proceedings of the 30th International Conference on Intelligent User Interfaces (IUI '25)*. Association for Computing Machinery, New York, NY, USA, 745–754. doi:10.1145/3708359.3712157
- [15] Francisco Maria Calisto, João Fernandes, Margarida Morais, Carlos Santiago, João Maria Abrantes, Nuno Nunes, and Jacinto C. Nascimento. 2023. Assertiveness-based Agent Communication for a Personalized Medicine on Medical Imaging Diagnosis. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (Hamburg, Germany) (CHI '23)*. Association for Computing Machinery, New York, NY, USA, Article 13, 20 pages. doi:10.1145/3544548.3580682
- [16] Centers for Medicare & Medicaid Services and National Center for Health Statistics. 2020. *ICD-10-CM Official Guidelines for Coding and Reporting FY 2021*. Technical Report. U.S. Department of Health and Human Services. <https://www.cms.gov/files/document/2021-coding-guidelines-updated-12162020.pdf> Effective October 1, 2020 through September 30, 2021.
- [17] Tirtha Chanda, Sarah Haggemueller, Tabea-Clara Bucher, Tim Holland-Letz, Harald Kittler, Philipp Tschandl, Markus V Heppt, Carola Berking, Jochen S Utikal, Bastian Schilling, et al. 2025. Dermatologist-like explainable AI enhances melanoma diagnosis accuracy:

- eye-tracking study. *Nature Communications* 16, 1 (2025), 4739.
- [18] Regina de Brito Duarte and Joana Campos. 2024. Looking For Cognitive Bias In AI-Assisted Decision-Making. In *HHAI-WS 2024: Workshops at the Third International Conference on Hybrid Human-Artificial Intelligence (CEUR Workshop Proceedings)*, Marina De Vos Petter Ericson, Nina Khairova (Ed.). CEUR-WS.org, Aachen, Germany, 10–21. <https://ceur-ws.org/Vol-3825/short1-1.pdf> Malmö, Sweden, June 10, 2024.
 - [19] Hans de Bruijn, Martijn Warnier, and Marijn Janssen. 2022. The perils and pitfalls of explainable AI: Strategies for explaining algorithmic decision-making. *Government Information Quarterly* 39, 2 (2022), 101666. doi:10.1016/j.giq.2021.101666
 - [20] Malin Eiband, Daniel Buschek, Alexander Kremer, and Heinrich Hussmann. 2019. The Impact of Placebic Explanations on Trust in Intelligent Systems. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (*CHI EA '19*). Association for Computing Machinery, New York, NY, USA, 1–6. doi:10.1145/3290607.3312787
 - [21] Daniel Fryer, Inga Strümke, and Hien Nguyen. 2021. Shapley Values for Feature Selection: The Good, the Bad, and the Axioms. *IEEE Access* 9 (2021), 144352–144360. doi:10.1109/ACCESS.2021.3119110
 - [22] Matthew R. Fulton II, Muhammad Zubair, and Sharven Taghavi. 2025. *Laboratory Evaluation of Sepsis*. StatPearls Publishing, Treasure Island (FL). <http://europepmc.org/books/NBK594258>
 - [23] Lahiru Gamage, Uditha Isuranga, Senuri De Silva, and Dulani Meedeniya. 2023. Melanoma skin cancer classification with explainability. In *2023 3rd International Conference on Advanced Research in Computing (ICARC)*. IEEE, Piscataway, NJ, USA, 30–35.
 - [24] Marzyeh Ghassemi, Luke Oakden-rayner, and Andrew L Beam. 2021. Viewpoint The false hope of current approaches to explainable artificial intelligence in health care. *The Lancet Digital Health* 3, 11 (2021), e745–e750. doi:10.1016/S2589-7500(21)00208-9
 - [25] Sukhpal Singh Gill, Minxian Xu, Carlo Ottaviani, Panos Patros, Rami Bahsoon, Arash Shaghaghi, Muhammed Golec, Vlado Stankovski, Huaming Wu, Ajith Abraham, Manmeet Singh, Harshit Mehta, Soumya K. Ghosh, Thar Baker, Ajith Kumar Parlikad, Hanan Lutfiyya, Salil S. Kanhere, Rizos Sakellariou, Shahram Dustdar, Omer Rana, Ivona Brandic, and Steve Uhlig. 2022. AI for next generation computing: Emerging trends and future directions. *Internet of Things (Netherlands)* 19, February (2022), 100514. doi:10.1016/j.iot.2022.100514 arXiv:2203.04159
 - [26] Gaole He, Nilay Aishwarya, and Ujwal Gadiraju. 2025. Is Conversational XAI All You Need? Human-AI Decision Making With a Conversational XAI Assistant. In *Proceedings of the 30th International Conference on Intelligent User Interfaces (IUI '25)*. Association for Computing Machinery, New York, NY, USA, 907–924. doi:10.1145/3708359.3712133
 - [27] Patrick Hemmer, Max Schemmer, Lara Rieflé, Nico Rosellen, Michael Vössing, and Niklas Köhl. 2022. Factors that influence the adoption of human-AI collaboration in clinical decision-making. arXiv:2204.09082 [cs.HC] <https://arxiv.org/abs/2204.09082>
 - [28] Kerstin Hogg, G. Brown, J. Dunning, J. Wright, S. Carley, B. Foex, and K. Mackway-Jones. 2006. Diagnosis of pulmonary embolism with CT pulmonary angiography: A systematic review. *Emergency Medicine Journal* 23, 3 (2006), 172–178. doi:10.1136/emj.2005.029397
 - [29] Maia Jacobs, Jeffrey He, Melanie F. Pradier, Barbara Lam, Andrew C. Ahn, Thomas H. McCoy, Roy H. Perlis, Finale Doshi-Velez, and Krzysztof Z. Gajos. 2021. Designing AI for Trust and Collaboration in Time-Constrained Medical Decisions: A Sociotechnical Lens. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (*CHI '21*). Association for Computing Machinery, New York, NY, USA, Article 659, 14 pages. doi:10.1145/3411764.3445385
 - [30] Daniel Kahneman. 2011. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, New York.
 - [31] Simarjeet Kaur, Jimmy Singla, Lewis Nkenyereye, Sudan Jha, Deepak Prashar, Gyanendra Prasad Joshi, Shaker El-Sappagh, Md Saiful Islam, and SM Riazul Islam. 2020. Medical diagnostic systems using artificial intelligence (AI) algorithms: principles and perspectives. *Ieee Access* 8 (2020), 228049–228069.
 - [32] Mohsen Khosravi, Zahra Zare, Seyyed Morteza Mojtabaiean, and Reyhane Izadi. 2024. Artificial intelligence and decision-making in healthcare: a thematic analysis of a systematic review of reviews. *Health services research and managerial epidemiology* 11 (2024), 23333928241234863.
 - [33] Nicholas Kroeger, Dan Ley, Satyapriya Krishna, Chirag Agarwal, and Himabindu Lakkaraju. 2023. Are Large Language Models Post Hoc Explainers? <https://openreview.net/forum?id=mAzHEP9jPv>
 - [34] Fatimah Lateef. 2010. Simulation-based learning: Just like the real thing. *Journal of emergencies, trauma, and shock* 3, 4 (2010), 348–352.
 - [35] Zhuoyan Li, Hangxiao Zhu, Zhuoran Lu, Ziang Xiao, and Ming Yin. 2025. From Text to Trust: Empowering AI-assisted Decision Making with Adaptive LLM-powered Analysis. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 1193, 18 pages. doi:10.1145/3706598.3713133
 - [36] Brian Y. Lim, Joseph P. Cahaly, Chester Y. F. Sng, and Adam Chew. 2025. Diagrammatization and Abduction to Improve AI Interpretability With Domain-Aligned Explanations for Medical Diagnosis. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 419, 25 pages. doi:10.1145/3706598.3714058
 - [37] Shuai Ma, Qiaoyi Chen, Xinru Wang, Chengbo Zheng, Zhenhui Peng, Ming Yin, and Xiaojuan Ma. 2025. Towards Human-AI Deliberation: Design and Evaluation of LLM-Empowered Deliberative AI for AI-Assisted Decision-Making. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 261, 23 pages. doi:10.1145/3706598.3713423

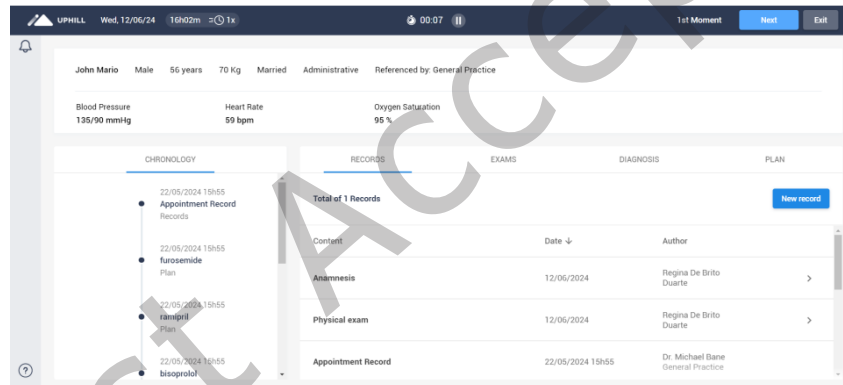
- [38] Nikita Mehandru, Brenda Y. Miao, Eduardo Rodriguez Almaraz, Madhumita Sushil, Atul J. Butte, and Ahmed Alaa. 2024. Evaluating large language models as agents in the clinic. *npj Digital Medicine* 7, 1 (2024), 17–19. doi:10.1038/s41746-024-01083-y
- [39] Myura Nagendran, Paul Festor, Matthieu Komorowski, Anthony C Gordon, and Aldo A Faisal. 2023. Quantifying the impact of AI recommendations with explanations on prescription decision making. *NPJ Digital Medicine* 6, 1 (2023), 206.
- [40] Mohammad Naiseh, Dena Al-Thani, Nan Jiang, and Raian Ali. 2023. How the different explanation classes impact trust calibration: The case of clinical decision support systems. *International Journal of Human Computer Studies* 169, September 2022 (2023), 102941. doi:10.1016/j.ijhcs.2022.102941
- [41] Van Bach Nguyen, Jörg Schlötterer, and Christin Seifert. 2023. From Black Boxes to Conversations: Incorporating XAI in a Conversational Agent. In *Explainable Artificial Intelligence*, Luca Longo (Ed.). Springer Nature Switzerland, Cham, 71–96.
- [42] Obinna O Oleribe, Jenny Momoh, Benjamin SC Uzochukwu, Francisco Mbofana, Akin Adebisi, Thomas Barbera, Roger Williams, and Simon D Taylor-Robinson. 2019. Identifying Key Challenges Facing Healthcare Systems In Africa And Potential Solutions. *International Journal of General Medicine* 12 (2019), 395–403. doi:10.2147/IJGM.S223882 PMID: 31819592.
- [43] Anneli Olsen and Ricardo Matos. 2012. Identifying parameter values for an I-VT fixation filter suitable for handling data sampled with various sampling frequencies. In *Proceedings of the Symposium on Eye Tracking Research and Applications* (Santa Barbara, California) (ETRA '12). Association for Computing Machinery, New York, NY, USA, 317–320. doi:10.1145/2168556.2168625
- [44] Cecilia Panigutti, Andrea Beretta, Fosca Giannotti, and Dino Pedreschi. 2022. Understanding the impact of explanations on advice-taking: a user study for AI-based clinical Decision Support Systems. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 568, 9 pages. doi:10.1145/3491102.3502104
- [45] Keith Rayner. 1998. Eye Movements in Reading and Information Processing: 20 Years of Research. *Psychological Bulletin* 124, 3 (1998), 372–422. doi:10.1037/0033-2909.124.3.372
- [46] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. 2016. "Why Should I Trust You?": Explaining the Predictions of Any Classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (San Francisco, California, USA) (KDD '16). Association for Computing Machinery, New York, NY, USA, 1135–1144. doi:10.1145/2939672.2939778
- [47] Ana Rita Rodrigues. 2021. Pulmonary embolism. <https://www.hospitaldaluz.pt/en/health-dictionary/pulmonary-embolism>
- [48] Emely Rosbach, Jonas Ammeling, Sebastian Krügel, Angelika Kießig, Alexis Fritz, Jonathan Ganz, Chloé Puget, Taryn Donovan, Andrea Klang, Maximilian C. Köller, Pompei Bolfa, Marco Tecilla, Daniela Denk, Matti Kiupel, Georgios Paraschou, Mun Keong Kok, Alexander F. H. Haake, Ronald R. de Krijger, Andreas F.-P. Sonnen, Tanit Kasantikul, Gerry M. Dorrestein, Rebecca C. Smedley, Nikolas Stathonikos, Matthias Uhl, Christof A. Bertram, Andreas Riener, and Marc Aubreville. 2025. "When Two Wrongs Don't Make a Right" - Examining Confirmation Bias and the Role of Time Pressure During Human-AI Collaboration in Computational Pathology. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). Association for Computing Machinery, New York, NY, USA, Article 528, 18 pages. doi:10.1145/3706598.3713319
- [49] Dario D. Salvucci and Joseph H. Goldberg. 2000. Identifying fixations and saccades in eye-tracking protocols. In *Proceedings of the 2000 Symposium on Eye Tracking Research & Applications* (Palm Beach Gardens, Florida, USA) (ETRA '00). Association for Computing Machinery, New York, NY, USA, 71–78. doi:10.1145/355017.355028
- [50] Wojciech Samek and Klaus-Robert Müller. 2019. *Towards Explainable Artificial Intelligence*. Springer International Publishing, Cham, 5–22. doi:10.1007/978-3-030-28954-6_1
- [51] Hua Shen, Chieh-Yang Huang, Tongshuang Wu, and Ting-Hao Kenneth Huang. 2023. ConvXAI: Delivering Heterogeneous AI Explanations via Conversations to Support Human-AI Scientific Writing. In *Companion Publication of the 2023 Conference on Computer Supported Cooperative Work and Social Computing* (Minneapolis, MN, USA) (CSCW '23 Companion). Association for Computing Machinery, New York, NY, USA, 384–387. doi:10.1145/3584931.3607492
- [52] Venkatesh Sivaraman, Leigh A Bukowski, Joel Levin, Jeremy M. Kahn, and Adam Perer. 2023. Ignore, Trust, or Negotiate: Understanding Clinician Acceptance of AI-Based Treatment Recommendations in Health Care. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 754, 18 pages. doi:10.1145/3544548.3581075
- [53] Dylan Slack, Satyapriya Krishna, Himabindu Lakkaraju, and Sameer Singh. 2023. Explaining machine learning models with interactive natural language conversations using TalkToModel. *Nature Machine Intelligence* 5, 8 (2023), 873–883.
- [54] Kacper Sokol and Peter Flach. 2020. One Explanation Does Not Fit All: The Promise of Interactive Explanations for Machine Learning Transparency. *KI - Künstliche Intelligenz* 34, 2 (2020), 235–250. doi:10.1007/s13218-020-00637-y arXiv:2001.09734
- [55] Harishankar V. Subramanian, Casey Canfield, and Daniel B. Shank. 2024. Designing explainable AI to improve human-AI team performance: A medical stakeholder-driven scoping review. *Artificial Intelligence in Medicine* 149, December 2023 (2024), 102780. doi:10.1016/j.artmed.2024.102780
- [56] Ashley Suh, Isabelle Hurley, Nora Smith, and Ho Chit Siu. 2025. Fewer Than 1% of Explainable AI Papers Validate Explainability with Humans. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (CHI EA '25). Association for Computing Machinery, New York, NY, USA, Article 276, 7 pages. doi:10.1145/3706599.3719964

- [57] Siddharth Swaroop, Zana Bućinca, Krzysztof Z. Gajos, and Finale Doshi-Velez. 2024. Accuracy-Time Tradeoffs in AI-Assisted Decision Making under Time Pressure. In *Proceedings of the 29th International Conference on Intelligent User Interfaces* (Greenville, SC, USA) (IUI '24). Association for Computing Machinery, New York, NY, USA, 138–154. doi:10.1145/3640543.3645206
- [58] Lana Tikhomirov, Carolyn Semmler, Melissa McCradden, Rachel Searston, Marzyeh Ghassemi, and Lauren Oakden-Rayner. 2024. Medical artificial intelligence for clinicians: the lost cognitive perspective. *The Lancet Digital Health* 6, 8 (2024), e589–e594.
- [59] Tim Miller. 2019. Explanation in artificial intelligence: Insights from the social sciences. 38 pages. <https://www.sciencedirect.com/science/article/pii/S0004370218305988>
- [60] Philipp Tschandl, Christoph Rinner, Zoe Apalla, Giuseppe Argenziano, Noel Codella, Allan Halpern, Monika Janda, Aimilios Lallas, Caterina Longo, Josep Malvehy, John Paoli, Susana Puig, Cliff Rosendahl, H. Peter Soyer, Iris Zalaudek, and Harald Kittler. 2020. Human–computer collaboration for skin cancer recognition. *Nature Medicine* 26, 8 (2020), 1229–1234. doi:10.1038/s41591-020-0942-0
- [61] Sara Wolf, Tobias Grundgeiger, Raphael Zähringer, Lora Shishkova, Franzisca Maas, Christina Dilling, and Oliver Happel. 2025. How a Clinical Decision Support System Changed the Diagnosis Process: Insights from an Experimental Mixed-Method Study in a Full-Scale Anesthesiology Simulation. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). Association for Computing Machinery, New York, NY, USA, Article 257, 23 pages. doi:10.1145/3706598.3713372
- [62] Wei Xu, Marvin J. Dainoff, Liezhong Ge, and Zaifeng Gao. 2021. Transitioning to Human Interaction with AI Systems: New Challenges and Opportunities for HCI Professionals to Enable Human-Centered AI. *International Journal of Human-Computer Interaction* 39, 3 (2021), 494–518. doi:10.1080/10447318.2022.2041900 arXiv:2105.05424
- [63] Jingfeng Yang, Hongye Jin, Ruixiang Tang, Xiaotian Han, Qizhang Feng, Haoming Jiang, Shaochen Zhong, Bing Yin, and Xia Hu. 2024. Harnessing the Power of LLMs in Practice: A Survey on ChatGPT and Beyond. *ACM Trans. Knowl. Discov. Data* 18, 6, Article 160 (April 2024), 32 pages. doi:10.1145/3649506
- [64] Hubert D. Zając, Dana Li, Xiang Dai, Jonathan F. Carlsen, Finn Kensing, and Tariq O. Andersen. 2023. Clinician-Facing AI in the Wild: Taking Stock of the Sociotechnical Challenges and Opportunities for HCI. *ACM Trans. Comput.-Hum. Interact.* 30, 2, Article 33 (March 2023), 39 pages. doi:10.1145/3582430
- [65] Yunfeng Zhang, Q. Vera Liao, and Rachel K. E. Bellamy. 2020. Effect of confidence and explanation on accuracy and trust calibration in AI-assisted decision making. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* (Barcelona, Spain) (FAT* '20). Association for Computing Machinery, New York, NY, USA, 295–305. doi:10.1145/3351095.3372852
- [66] Ruiqiang Zheng, Yifen Zhang, Ziqi Rong, Wei Huang, and Xiaoyun Fu. 2021. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock 2021, interpretation and expectation. *Zhonghua Wei Zhong Bing Ji Jiu Yi Xue* 33, 10 (2021), 1159–1164.
- [67] Alexandra Zytek, Sara Pidò, and Kalyan Veeramachaneni. 2024. LLMs for XAI: Future Directions for Explaining Explanations. arXiv:2405.06064 [cs.AI] <https://arxiv.org/abs/2405.06064>

A Uphill Simulator



(a) Initialisation of the Case

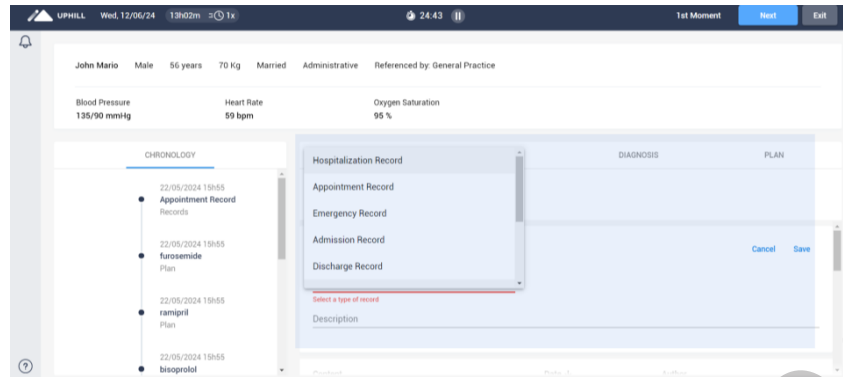


(b) Main Interface

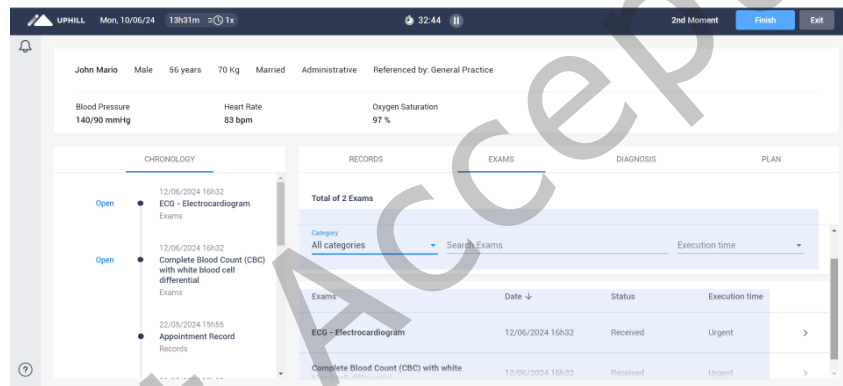
Fig. 6. Uphill Simulator Features to Solve Clinical Cases (Part 1)

UpHill is a Portuguese startup that operates in the healthcare sector by developing software and medical content to automate healthcare journeys in hospitals and other institutions. Among various services, *UpHill* offers the Simulate, a clinical simulation tool similar to a real clinical decision support system (CDSS). This platform allows the development and resolution of clinical cases in a completely digital and gamified way. Below, we will explain the main features of this platform.

To initiate the case study, the simulation presents detailed information about the patient, including appointment records, medical history, or disease screening, just as shown in 6a. The main interface is divided into three main sections: at the top, there are general information about the patient along with vital signs; on the left hand side,



(c) Record Section

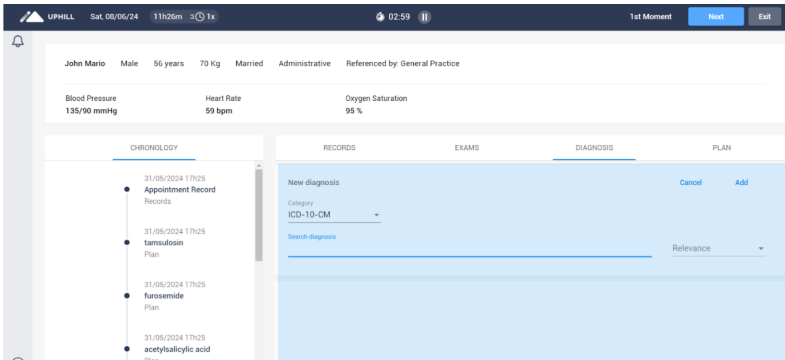


(d) Exam Section

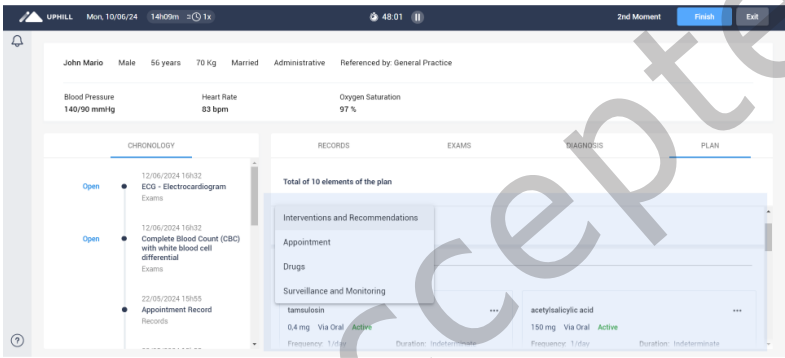
Fig. 6. UpHill Simulator Features to Solve Clinical Cases (Part 2)

there is a chronology that displays everything previously done with the patient, such as medication prescriptions, past appointment records, and prior exams. On the right side, there is an action area divided into four main categories: records, exams, diagnosis, and treatment plan, as can be seen in 6b. We will dive into the *UpHill* possible actions with more detail:

- **Records:** Using the UpHill Simulator, doctors can add various records related to the patient, such as appointment records, emergency records, and clinical summaries, the information highlighted in Figure 6c. This section also allows users to view the anamnesis, a comprehensive interview that gathers the patient's medical history and symptoms, as well as the physical exam summary, which assesses the body for signs of illness or abnormalities.



(e) Diagnosis Section



(f) Plan Section



(g) Feedback Section

Fig. 6. Uphill Simulator Features to Solve Clinical Cases (Part 3)

- **Exams:** Users can request as many exams for the patient as needed. The exams are categorised into various sections: Laboratory, Clinical Pathology, Imaging, Special Cardiology Exams, Microbiology, Pulmonary Function Tests, and Neurophysiology Exams. The results, if available in the system, are displayed, as illustrated in Figure 6d.
- **Diagnosis:** Physicians can add the diagnosis conducted to the patient according to the ICD-10-CM, used to code and classify medical diagnoses, as shown in Figure 6e, [16].
- **Plan:** There is a section dedicated to treatment planning, where doctors can review the current plans prescribed to the patient. It includes a tool that allows doctors to create a new plan based on their clinical assessment findings. These plans can be organised into different categories such as interventions and recommendations, appointments, drugs prescription and monitoring, as illustrated in Figure 6f.

After the user completes the clinical case resolution, a feedback page is displayed. This page provides a score based on the user's actions and their accuracy against the expected resolution of the case study. Each expected action has associated points and includes an explanation of the rationale behind it, in the final report. This information can be seen in Figure 6g.

B Medical full description of the clinical cases

B.1 PTE case

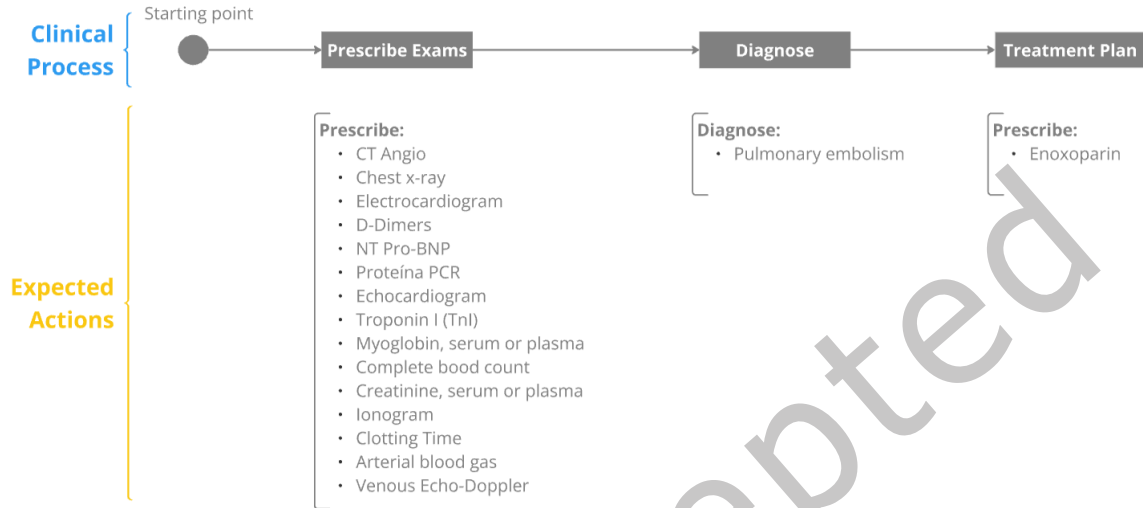


Fig. 7. PTE - Case Resolution

Medical history: This case is about a 75 year old woman with a history of hypertension and a femoral neck fracture from a month ago, underwent total hip replacement. Her recovery was complicated by nosocomial pneumonia, and she is currently in motor rehabilitation at a convalescence unit. She presented to the emergency department with sudden dyspnea that began that morning, without associated chest pain. Her usual medication is Ramipril 5mg daily.

Diagnosis: This case is indicative of a Pulmonary Thromboembolism, as suggested by the patient's acute shortness of breath, recent hip surgery, and symptoms of a swollen, tender leg. The key diagnostic exams for suspected pulmonary thromboembolism include an ECG, chest X-ray, specific laboratory tests such as D-dimers, perfusion scintigraphy, or Angio-CT, [47].

The most accurate tests for detecting Pulmonary Embolism (PE) are the D-dimer test and Angio-CT. Angio-CT, uses contrast dye to highlight blockages in arteries and veins, enabling precise diagnosis and assessment of thromboembolic conditions. This imaging technique also helps differentiate pulmonary embolism from other conditions such as pneumonia or aortic dissection, [28]. D-dimer test is a blood test that checks for blood-clotting problems. In those cases its levels are elevated in plasma in the presence of acute thrombosis due to the simultaneous activation of coagulation and fibrinolysis. While Angio-CT is highly accurate, it is more costly and invasive compared to the D-dimer test. According to guidelines from the European Respiratory Society and the European Society of Cardiology, for patients with low or intermediate clinical probability of PE, a D-dimer test should be conducted first. If the D-dimer test is positive, an Angio-CT should follow to confirm the diagnosis. For patients with high clinical probability or likely (PE), an immediate Angio-CT is recommended. This case falls into the latter category.

The standard treatment plan involves the administration of anticoagulant drugs, such as heparin, which helps to prevent the growth of existing blood clots and the formation of new ones.

Case Resolution: The *UpHill* resolution of this case study, presented in figure 7, covers the main actions as the most correct steps to take according to the current guidelines. This case is considered relatively low in difficulty because diagnosing the pulmonary embolism is straightforward based on the medical history. Once the diagnosis is suspected, determining the necessary exams and treatment plan is also simple.

B.2 PAC case

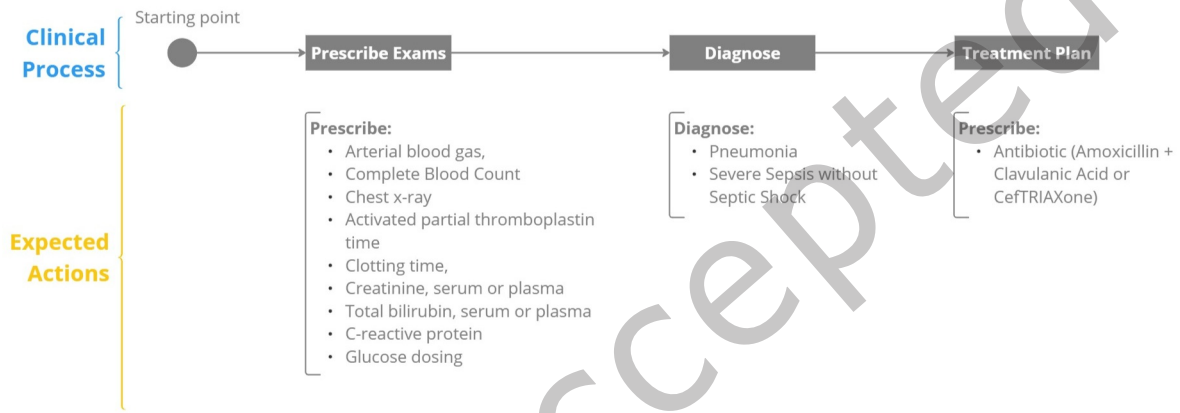


Fig. 8. PAC - Case Resolution

Medical history: This case is about an 87 year old partially dependent man, that has a medical history of hypertension, treated with 5 mg of ramipril. The patient presented to the emergency department with a 7 day history of mucopurulent cough and breathing difficulties, along with pleuritic chest pain that has got worse in the past 24 hours. Upon triage, he was found to have dyspnea, cough, and chest pain. The patient exhibits signs of severe infection, highlighted by the pallor noted on mucocutaneous examination and he is also confused and disoriented confusion which led to not being possible to conduct anamnesis.

Diagnosis: This case involves sepsis without septic shock, caused by bacterial pneumonia. Bacterial pneumonia is one of the most common sources of infection in sepsis. In the absence of an obvious source of infection, chest radiography is often required to identify pneumonia. Key diagnostic exams include a comprehensive laboratory workup, such as a complete blood count, chemistry panel, liver function tests, and biomarkers like C-reactive protein, lactate, and procalcitonin, [22]. These tests are crucial for diagnosing sepsis, evaluating its severity (with or without septic shock), and determining prognosis. One key indicator in the Arterial Blood Gas test is lactate, a biomarker for tissue hypoperfusion and subsequent organ dysfunction. Elevated lactate levels, or hyperlactacidemia, are a predictor of mortality [66]. Sepsis is a leading cause of mortality worldwide. The treatment plan depends on factors such as the severity of the condition and the source of the infection. Early administration of antimicrobial therapy is one of the most effective interventions, significantly reducing in-hospital mortality.

Antibiotic treatment is typically a critical component of sepsis management.

Case Resolution: The *UpHill* resolution of this case study, presented in figure 8, covers the main actions as the most correct steps to take according to the current guidelines. This case is considered to be more challenging compared to PTE's, as the diagnosis of sepsis is not as straightforward and can easily be mistaken for pneumonia only.

Just Accepted

C Results Data

	PTE		PAC	
	Mean	SE	Mean	SE
Total Exams Prescribed				
Control	6.27	0.648	5.91	0.899
AI	6.67	1.150	8.42	1.520
XAI	6.18	0.736	8.27	1.685
F-Score (2,31)	0.085		0.974	
p-value	0.918		0.389	
Necessary Exams				
Control	4.64	0.527	3.36	0.388
AI	4.58	0.621	3.67	0.414
XAI	4.45	0.608	3.36	0.364
F-Score (2,31)	0.024		0.206	
p-value	0.976		0.815	
Unnecessary Exams				
Control	1.64	0.279	2.55	0.767
AI	2.08	0.783	4.75	1.719
XAI	1.73	0.506	4.91	1.965
F-Score (2,31)	0.171		0.684	
p-value	0.844		0.512	
Exam Score				
Control	0.321	0.031	0.249	0.021
AI	0.457	0.031	0.276	0.014
XAI	0.408	0.052	0.259	0.019
F-Score (2,31)	3.135		0.560	
p-value	0.058		0.577	
Clinical Case Score				
Control	0.313	0.041	0.273	0.030
AI	0.450	0.039	0.313	0.012
XAI	0.431	0.046	0.369	0.023
F-Score (2,31)	3.108		4.517	
p-value	0.059		0.019*	
Reliance in Exam Recommendation				
Control	0.309	0.073	0.509	0.062
AI	0.417	0.057	0.533	0.057
XAI	0.509	0.073	0.582	0.063
F-Score (2,31)	2.116		0.363	
p-value	0.138		0.699	
Case Resolution Time				
Control	20.07	2.242	19.52	1.922
AI	15.34	1.514	16.639	2.602
XAI	13.97	1.330	15.752	1.815
F-Score (2,31)	3.350		0.803	
p-value	0.048*		0.457	

Table 8. ANOVA table with Mean, Standard Error, F-Scores, and p-values for all metrics.

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